ESP32 -HARNESS

ADVANCED PENTESTING, AUDIT & TELEMETRY FIRMWARE

v1.0 MIT

CONTROLLER	**ESP32-WROOM-32** Dual-core 240 MHz, 520 KB SRAM
RF TRANSCEIVER	**NRF24L01+ PA/LNA** 2.4 GHz ISM, +20 dBm, 250 kbps - 2 Mbps
STORAGE	**MicroSD Module** FAT32/exFAT, SPI, up to 32 GB
DISPLAY	**OLED SSD1306** 128x64 monochrome, I2C
BATTERY	**LiPo 3.7V 1000mAh** USB-C charging, ~4 hours runtime

CAPABILITIES

Wi-Fi Research	802.11 frame capture, deauth monitoring, probe tracking
RF24 Spectrum	ISM band analysis, channel mapping, signal visualization
Packet Injection	Controlled RF transmission for protocol testing
BLE Enumeration	Bluetooth Low Energy device discovery
Forensic Capture	Full PCAP to SD, UTC timestamps, Wireshark export
GPIO Events	Hardware sensors, MQTT/HTTP telemetry

2.4

GHz

SYSTEM ARCHITECTURE

ESP32-WROOM-32
Main controller

Core Framework
CLI, PCAP Engine, Logger, Config

Wi-Fi Engine
Scan, Capture, Deauth detect

RF24 Engine
NRF24L01+ spectrum & injection

BLE Engine
Device enumeration & scanning

Telemetry Output
MQTT, HTTP, Serial

USAGE FLOW

01 Flash firmware t
02 Load config (Wi-
03 Select target net
04 Capture packets
05 Export PCAP for

COMPONENT MATRIX

COMPONENT	INTERFACE	PROTOCOL
ESP32-WROOM-32	UART / USB	Serial 115200
NRF24L01+ PA/LNA	SPI	SPI 10 MHz
MicroSD Module	SPI	SPI 25 MHz
OLED SSD1306	I2C	I2C 400 kHz
LiPo Battery	ADC	3.7V nominal